EXERCISE 7

Nonlinear Change of Variables

Original statement · ProblemsCh2.pdf, p. 2

Substitution z = y² • Linear reduction

PROBLEM STATEMENT

We consider the following initial value problem:

$$y'(x) = y(x) + \frac{x}{y(x)}, \qquad y(0) = 1.$$

Make the change of variable $z(x) = y(x)^2$ and find the solution.

Source: ProblemsCh2.pdf, Exercise 7 (page 2).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Right-hand side: $f(x, y) = y + x/y$. It is continuous for $y \neq 0$, and $\partial f/\partial y = 1 - x/y^2$ is continuous for $y \neq 0$. The initial point $(0, 1)$ lies in the region $y > 0$, so Theorem 6.2 gives a unique local solution. The line $y = 0$ is excluded from the domain, so the solution cannot cross it: while it exists it keeps the sign of $y(0) = 1$, that is $y(x) > 0$.

The equation is non-linear in y because of the term $x\,y^{-1}$. It has Bernoulli form $y' - y = x\,y^{-1}$ with exponent $\alpha = -1$ (see Problem 2.8).

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this substitution: multiplying the equation by y gives $y\,y' = y^2 + x$, and the left side is $\frac{1}{2}\frac{d}{dx}(y^2)$. In terms of $z = y^2$ the right side $y^2 + x = z + x$ is linear, so the problem becomes a linear first-order equation, solvable with an integrating factor.

Change of variables written explicitly: $z = y^2,\ z' = 2y\,y'$ (chain rule), $z(0) = y(0)^2 = 1$. Back-substitution: since $y > 0$, $y = +\sqrt{z}$.

CLASS NOTES AND COMPLEMENT [NOTES, P. 3]

The substitution $z = y^2$ is a complement (not in the class notes). The linear equation it produces, $z' - 2z = 2x$, $z(0) = 1$, is solved in Step 2 below with the three-step algorithm of the notes (μ there is the notes' I): Step 1, $P(x) = \int_0^x (-2)\,ds = -2x$, $I(x) = e^{-2x}$, $I(0) = 1$; Step 2, $\int_0^x (Iz)'\,ds = \int_0^x 2s\,e^{-2s}\,ds$; Step 3, $z(x) = \frac{z(0)}{I(x)} + \frac{1}{I(x)}\int_0^x I(s)\,2s\,ds$.

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: transform the equation

Multiply $y' = y + x/y$ by $2y$ (allowed since $y \neq 0$):

$$2y\,y' = 2y^2 + 2x.$$

With $z = y^2$ and $z' = 2yy'$:

$$z' = 2z + 2x, \qquad z(0) = 1, \qquad \text{i.e.} \quad z' - 2z = 2x.$$

Step 2: integrating factor

$p(x) = -2$, so $\int p\,dx = -2x$ and $\mu(x) = e^{-2x}$, $\mu(0) = 1$. Then $(z\,e^{-2x})' = 2x\,e^{-2x}$. Integrate from 0 to x (Barrow):

$$z(x)e^{-2x} - z(0)\cdot 1 = \int_0^x 2s\,e^{-2s}\,ds.$$

Integration by parts with $u = 2s$, $dv = e^{-2s}ds$, $du = 2\,ds$, $v = -\frac{1}{2}e^{-2s}$:

$$\int 2s\,e^{-2s}\,ds = -s\,e^{-2s} + \int e^{-2s}\,ds = -s\,e^{-2s} - \frac{1}{2}e^{-2s}.$$

Evaluate between the limits 0 and x:

$$\left[-s\,e^{-2s} - \tfrac{1}{2}e^{-2s}\right]_0^x = -x\,e^{-2x} - \tfrac{1}{2}e^{-2x} + \tfrac{1}{2}.$$

Hence $z\,e^{-2x} = 1 - x\,e^{-2x} - \frac{1}{2}e^{-2x} + \frac{1}{2} = \frac{3}{2} - x\,e^{-2x} - \frac{1}{2}e^{-2x}$. Multiply by e^{2x}:

$$z(x) = \frac{3}{2}e^{2x} - x - \frac{1}{2}.$$

Step 3: return to y and find the domain

Since $y > 0$ and $y^2 = z$, we need $z > 0$ and take the positive root:

$$y(x) = \sqrt{\frac{3}{2}e^{2x} - x - \frac{1}{2}}.$$

Domain: is $z(x) > 0$ for all x? Minimize z: $z'(x) = 3e^{2x} - 1 = 0$ at $e^{2x} = 1/3$, i.e. $x_m = -\frac{1}{2}\ln 3 \approx -0.549$. Since $z'' = 6e^{2x} > 0$ this is the global minimum, and

$$z(x_m) = \frac{3}{2}\cdot\frac{1}{3} + \frac{1}{2}\ln 3 - \frac{1}{2} = \frac{1}{2}\ln 3 \approx 0.549 > 0.$$

So $z(x) \geq \frac{1}{2}\ln 3 > 0$ for every real x: the solution never reaches $y = 0$ and is defined on all of $\mathbb{R}$ (the maximal interval of existence is $\mathbb{R}$).

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Substitute back into the original equation. Differentiate $y = \sqrt{z}$: $y' = \dfrac{z'}{2y} = \dfrac{3e^{2x} - 1}{2y}$. The right-hand side is

$$y + \frac{x}{y} = \frac{y^2 + x}{y} = \frac{z + x}{y} = \frac{\frac{3}{2}e^{2x} - \frac{1}{2}}{y} = \frac{3e^{2x} - 1}{2y}.$$

They agree. Initial condition: $z(0) = \frac{3}{2} - 0 - \frac{1}{2} = 1$, so $y(0) = 1$.

- As $x \to +\infty$: $y \sim \sqrt{3/2}\, e^{x}$, exponential growth, consistent with $y' \approx y$ for large y.
- As $x \to -\infty$: $z \sim -x$, so $y \sim \sqrt{|x|} \to \infty$: the term x/y is negative for $x < 0$, while $z \approx -x$ grows.

Numeric check (fourth-order Runge-Kutta against the formula): the maximum absolute difference at $x = -2, -1, 1, 2$ is about 4.5×10^{-13} (fourth-order Runge-Kutta, 40000 steps per interval).

RESULT

$$y(x) = \sqrt{\tfrac{3}{2}e^{2x} - x - \tfrac{1}{2}}, \qquad x \in \mathbb{R}.$$